higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

(8080205)

18 August 2026 (X-paper)
09:00–12:00

Drawing instruments and nonprogrammable calculators may be used.

This question paper consists of 6 pages and 1 formula sheet.

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N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL INSTRUMENTS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a new page.
5. Use only a black or blue pen.
6. Write neatly and legibly.

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SECTION A

QUESTION 1: FLOW MEASUREMENT

1.1 Define the term, *compressible fluids.* (2)

1.2 List any THREE variable head type flowmeters. (3)

1.3 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 A rotameter is a …

A variable-head type flowmeter.
B positive displacement type flowmeter.
C variable area type flowmeter.
D None of the above.

1.3.2 Mass flow meters are based on the principle of …

A electromagnetic induction.
B piezoelectric effect.
C angular momentum.
D Newton's second law.

1.3.3 Positive displacement flow meters are generally preferred for …

A abrasive fluids.
B clean fluids.
C household water meters.
D None of the above.

1.3.4 When a rotameter has no flow, the float rests:

A At the top of the meter tube
B In the middle of the meter tube
C At the bottom of the meter tube
D Anyone position.

1.3.5 Quantity flowmeters are also called …

A cubic meters.
B area meters.
C acceleration meters.
D volumetric meters.

(5 × 1) (5)

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1.4 Make a neat, labelled sketch of an ultrasonic flow meter. (5)

1.5 Name the piezo crystals used as transmitters to send signal through the fluid flowing through the meter in QUESTION 1.4. (2)
[17]

QUESTION 2

2.1 What is the purpose of open channel flow measurement? (2)

2.2 Complete the following sentences regarding inferential flow meters by filling in the missing word or words. Write only the word or words next to the question number (2.2.1–2.2.4) in the ANSWER BOOK.

In these meters, the moving element is a (2.2.1) … which is generally designed to (2.2.2) … at a speed proportional to the (2.2.3) … of the flow through the (2.2.4) … . (4)

2.3 A typical rectangular weir with a channel base width of 1,2 m is used in a small river. The head is 1,5 m.

Calculate the flow rate in m^3/s. (4)

2.4 Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–D) next to the question number (2.4.1–2.4.4) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.4.1	Flow nozzle	A	direct indicating type
2.4.2	Rotameter	B	usable for all types of flow with good accuracy
2.4.3	Nutating disc	C	cheaper than venturi, more accurate than orifice
2.4.4	Reciprocating piston	D	common with liquid, less accurate

(4 × 1) (4)
[14]

TOTAL SECTION A: 31

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SECTION B

QUESTION 3: DENSITY, HUMIDITY AND VISCOSITY MEASUREMENT

3.1 Define Stokes' law. (2)

3.2 Make a neat, labelled sketch of the falling ball viscometer. (6)

3.3 Use sketches to explain Newtonian and non-Newtonian fluids. (6)

3.4 Name SIX different types of hygrometers. (6)

3.5 Make a neat, labelled sketch of a hand-held hydrometer. (3)

3.6 What is the dew cell? (2)

3.7 What are common units for measuring viscosity? (2)
[27]

QUESTION 4: PH AND CONDUCTIVITY MEASUREMENT

4.1 Calculate the $[H^+]$ and $[OH^-]$ if the pH of a solution is 9,79. (6)

4.2 Conductivity measurement systems in which simple Ohm meters are often inaccuracy due to variations in line voltage.

How would one overcome this disadvantage? (2)

4.3 How would one increase sensitivity and reduce non-linearity caused by meter resistance? (2)

4.4 State FOUR requirements of ideal housing material for electrodes. (4)
[14]

QUESTION 5: AUTO CONTROL MEASUREMENT

5.1 Define the following terms:

5.1.1 Gain

5.1.2 Drift

5.1.3 Integral action

5.1.4 Derivative action

(4 × 2) (8)

5.2 What is the function of integral controller action should the offset not be tolerated? (2)

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5.3 Make a neat, labelled block diagram of a closed-loop controller, showing feedback loop, discrimination, controller, control element and plant process. (7)

5.4 Name THREE types of output signals transmitted by a transmitter. (3)

5.5 State the main disadvantage of a proportional controller. (2)

5.6 Name FOUR forms of discrimination in electronic control. (4)

5.7 Name TWO types of pneumatic amplifiers. (2)
[28]

TOTAL SECTION B: 69
GRAND TOTAL: 100

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INDUSTRIAL INSTRUMENTS N5

FORMULA SHEET

$$Q = \frac{8}{15} \; Tan \frac{\theta}{2} \sqrt{2g.H^5}$$

$$Q = \frac{2}{3} \; B\,C\sqrt{2g.H^3}$$

$$Q = \frac{2}{3} \; CB\sqrt{2g.H^3}$$

pH = -log[H^+]

pOH = -log [OH^-]

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